

Experiment No. 2

Student Name: Ojaswini Thote

PRN: 24070521048

File Path: `file:///Users/ojaswinithote/Desktop/Javascript/EXP%202/index2.html`

Experiment Title

Demonstration of var, let, const, Template Literals, Destructuring and Billing Calculator using JavaScript

Software / Tools Required

- Visual Studio Code
- Google Chrome
- HTML5
- CSS3
- JavaScript (ES6)

Experiment Program Code

index.html — ShopEasy Online Shopping Website

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>ShopEasy - Online Shopping</title>

<style>

*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family:'Segoe UI',sans-serif;
}

body{
  background:#eef3f8;
}

header{
  background:linear-gradient(90deg,#1565C0,#42A5F5);
  color:white;
  text-align:center;
  padding:30px;
  box-shadow:0 4px 10px rgba(0,0,0,.2);
}

header h1{
  font-size:38px;
```

```

}

header p{
  margin-top:10px;
  font-size:18px;
}

.container{
  width:90%;
  max-width:1200px;
  margin:30px auto;
}

.section{
  background:white;
  padding:25px;
  margin-bottom:30px;
  border-radius:12px;
  box-shadow:0 5px 15px rgba(0,0,0,.15);
}

.section h2{
  color:#1565C0;
  margin-bottom:20px;
}

.form-grid{
  display:grid;
  grid-template-columns:1fr 1fr;
  gap:20px;
}

label{
  display:block;
  margin-bottom:8px;
  font-weight:bold;
}

input,
select{
  width:100%;
  padding:12px;
  border:1px solid #ccc;
  border-radius:8px;
  font-size:16px;
}

.products{
  display:grid;
  grid-template-columns:repeat(auto-fit,minmax(220px,1fr));
  gap:20px;
}

.card{
  background:#fafafa;
  border-radius:10px;
  padding:20px;
  text-align:center;
  box-shadow:0 3px 12px rgba(0,0,0,.12);
  transition:.3s;
}

.card:hover{
  transform:translateY(-5px);
}

.card h3{
  color:#1565C0;
}

.price{
  font-size:22px;
  color:#2E7D32;
}

```

```

        margin:15px 0;
        font-weight:bold;
    }

    .qty{
        width:80px;
        text-align:center;
    }

    button{
        width:100%;
        padding:15px;
        margin-top:30px;
        background:#2E7D32;
        color:white;
        border:none;
        border-radius:8px;
        cursor:pointer;
        font-size:18px;
    }

    button:hover{
        background:#1B5E20;
    }

    #bill{
        display:none;
        background:#F1F8E9;
        padding:20px;
        border-radius:10px;
    }

    table{
        width:100%;
        border-collapse:collapse;
        margin-top:20px;
    }

    th{
        background:#1565C0;
        color:white;
    }

    th,td{
        border:1px solid #ddd;
        padding:12px;
        text-align:center;
    }

    footer{
        background:#1565C0;
        color:white;
        text-align:center;
        padding:20px;
        margin-top:30px;
    }

    @media(max-width:768px){

        .form-grid{
            grid-template-columns:1fr;
        }

        .products{
            grid-template-columns:1fr;
        }

    }

</style>
</head>

```

```
<body>

<header>

<h1>☐ ShopEasy Online Shopping</h1>


</header>

<div class="container">

<div class="section">

<h2>Customer Details</h2>

<div class="form-grid">

<div>

<label>Name</label>

<input type="text"
id="name"
placeholder="Enter Customer Name">

</div>

<div>

<label>Phone Number</label>

<input type="text"
id="phone"
placeholder="Enter Phone Number">

</div>

<div>

<label>Prime Member</label>

<select id="member">

<option value="true">Yes</option>
<option value="false">No</option>

</select>

</div>

<div>

<label>Payment Method</label>

<select id="payment">

<option>Cash on Delivery</option>
<option>UPI</option>
<option>Credit Card</option>
<option>Debit Card</option>
<option>Net Banking</option>

</select>

</div>

</div>

</div>

</div>

<div class="section">
```

```
<h2>Products</h2>

<div class="products">

  <div class="card">

    <h3>Laptop</h3>

    <p class="price">₹50,000</p>

    <label>Quantity</label>

    <input
      type="number"
      class="qty"
      data-product="Laptop"
      data-price="50000"
      value="0"
      min="0">

  </div>

  <div class="card">

    <h3>Smartphone</h3>

    <p class="price">₹25,000</p>

    <label>Quantity</label>

    <input
      type="number"
      class="qty"
      data-product="Smartphone"
      data-price="25000"
      value="0"
      min="0">

  </div>

  <div class="card">

    <h3>Tablet</h3>

    <p class="price">₹18,000</p>

    <label>Quantity</label>

    <input
      type="number"
      class="qty"
      data-product="Tablet"
      data-price="18000"
      value="0"
      min="0">

  </div>

  <div class="card">

    <h3>Smart Watch</h3>

    <p class="price">₹5,000</p>

    <label>Quantity</label>

    <input
      type="number"
      class="qty"
      data-product="Smart Watch"
      data-price="5000"
      value="0"
      min="0">
```

```
</div>

<div class="card">

<h3>Headphones</h3>

<p class="price">₹2,000</p>

<label>Quantity</label>

<input
type="number"
class="qty"
data-product="Headphones"
data-price="2000"
value="0"
min="0">

</div>

<div class="card">

<h3>Keyboard</h3>

<p class="price">₹1,500</p>

<label>Quantity</label>

<input
type="number"
class="qty"
data-product="Keyboard"
data-price="1500"
value="0"
min="0">

</div>

<div class="card">

<h3>Mouse</h3>

<p class="price">₹700</p>

<label>Quantity</label>

<input
type="number"
class="qty"
data-product="Mouse"
data-price="700"
value="0"
min="0">

</div>

<div class="card">

<h3>Bluetooth Speaker</h3>

<p class="price">₹3,500</p>

<label>Quantity</label>

<input
type="number"
class="qty"
data-product="Bluetooth Speaker"
data-price="3500"
value="0"
min="0">
```

```
</div>

<div class="card">

<h3>Power Bank</h3>

<p class="price">₹1,200</p>

<label>Quantity</label>

<input
type="number"
class="qty"
data-product="Power Bank"
data-price="1200"
value="0"
min="0">

</div>

<div class="card">

<h3>USB Drive</h3>

<p class="price">₹800</p>

<label>Quantity</label>

<input
type="number"
class="qty"
data-product="USB Drive"
data-price="800"
value="0"
min="0">

</div>

</div>

<button onclick="generateBill()">
  ☐ Generate Bill
</button>

</div>

<div class="section">

<div id="bill">

<h2>Bill Receipt</h2>

<p><strong>Customer Name:</strong>
<span id="customer"></span></p>

<p><strong>Phone Number:</strong>
<span id="mobile"></span></p>

<p><strong>Prime Member:</strong>
<span id="prime"></span></p>

<p><strong>Payment Method:</strong>
<span id="paymentMethod"></span></p>

<table>

<thead>

<tr>

<th>Product</th>
<th>Price (₹)</th>
<th>Quantity</th>
```

```

<th>Total (₹)</th>

</tr>

</thead>

<tbody id="billItems">

</tbody>

</table>

<h3 id="subtotal"></h3>

<h3 id="gst"></h3>

<h2 id="grandTotal"></h2>

</div>

</div>

</div>

<script src="script2.js"></script>

</body>
</html>

```

The `index.html` file creates an online shopping webpage where customers can:

- Enter customer details (Name, Phone Number, Prime Membership, Payment Method).
- Select quantities for different electronic products.
- Generate a detailed bill with GST calculation.
- View a responsive shopping interface designed using HTML and CSS.

The webpage uses an external JavaScript file (`script2.js`) to perform billing calculations and display the final invoice.

script2.js — JavaScript (var, let, const, Template Literals and Destructuring)

```

function generateBill() {

    // String
    let customerName = document.getElementById("name").value.trim();

    // String
    let phone = document.getElementById("phone").value.trim();

    // Boolean
    let isPrime = document.getElementById("member").value === "true";

    // String
    let payment = document.getElementById("payment").value;

    // Validation
    if (customerName === "" || phone === "") {
        alert("Please enter customer details.");
        return;
    }
}

```



```

// Array
let purchasedItems = [];

// Number
let subtotal = 0;

// Get all quantity inputs
let qtyInputs = document.querySelectorAll(".qty");

qtyInputs.forEach(item => {

    let quantity = Number(item.value);

    if (quantity > 0) {

        let product = item.dataset.product;
        let price = Number(item.dataset.price);

        let amount = price * quantity;

        subtotal += amount;

        purchasedItems.push({
            product: product,
            price: price,
            quantity: quantity,
            amount: amount
        });

    }

});

if (purchasedItems.length === 0) {
    alert("Please select at least one product.");
    return;
}

// Number
let gst = subtotal * 0.18;

let grandTotal = subtotal + gst;

// Prime Member Discount
if (isPrime) {
    grandTotal -= 500;
}

// Object
let customer = {
    name: customerName,
    phone: phone,
    paymentMethod: payment,
    primeMember: isPrime
};

// Console Output (Data Types)
console.clear();

console.log("==== JavaScript Data Types =====");
console.log("String :", customerName);
console.log("Number :", grandTotal);
console.log("Boolean :", isPrime);
console.log("Array :", purchasedItems);

```

```

console.log("Object :", customer);

// Display Bill
document.getElementById("bill").style.display = "block";

document.getElementById("customer").textContent = customerName;
document.getElementById("mobile").textContent = phone;
document.getElementById("prime").textContent = isPrime ? "Yes" : "No";
document.getElementById("paymentMethod").textContent = payment;

let billTable = document.getElementById("billItems");
billTable.innerHTML = "";

purchasedItems.forEach(item => {

    billTable.innerHTML += `
        <tr>
            <td>${item.product}</td>
            <td>₹${item.price}</td>
            <td>${item.quantity}</td>
            <td>₹${item.amount}</td>
        </tr>
    `;

});

document.getElementById("subtotal").innerHTML =
    "Subtotal : ₹" + subtotal.toFixed(2);

document.getElementById("gst").innerHTML =
    "GST (18%) : ₹" + gst.toFixed(2);

document.getElementById("grandTotal").innerHTML =
    "Grand Total : ₹" + grandTotal.toFixed(2);

alert("☐ Bill Generated Successfully!");
}

```

The JavaScript file demonstrates modern ES6 concepts including:

- **var** for variables and loop counters.
- **let** for block-scoped variables.
- **const** for fixed values such as GST rate and calculated totals.
- **Template Literals** for generating the bill dynamically.
- **Destructuring** for extracting product details while generating the invoice.

It also performs:

- Product selection
- Quantity calculation
- GST (18%) calculation
- Grand Total calculation
- Dynamic bill generation
- Console logging of billing details

Features Implemented

- Online Shopping Interface
- Customer Details Form
- Product Quantity Selection
- Billing Calculator
- GST Calculation (18%)
- Grand Total Calculation
- Dynamic Invoice Generation
- JavaScript ES6 Features
- Console Output

Output

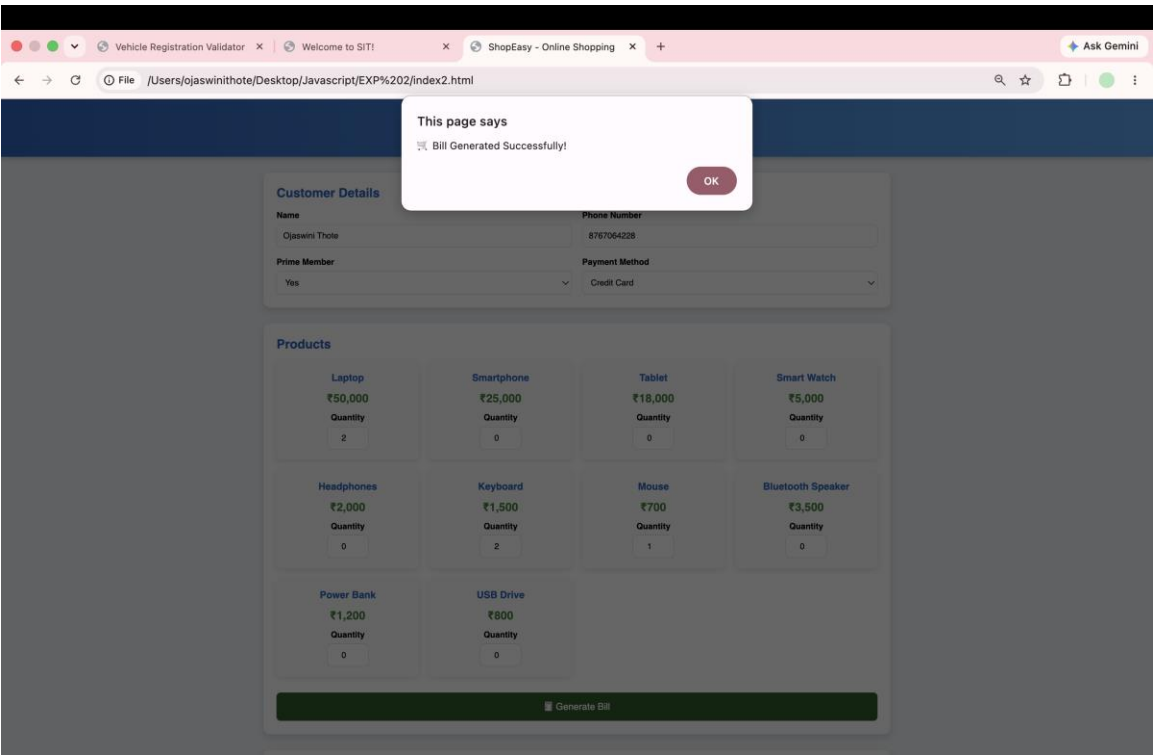
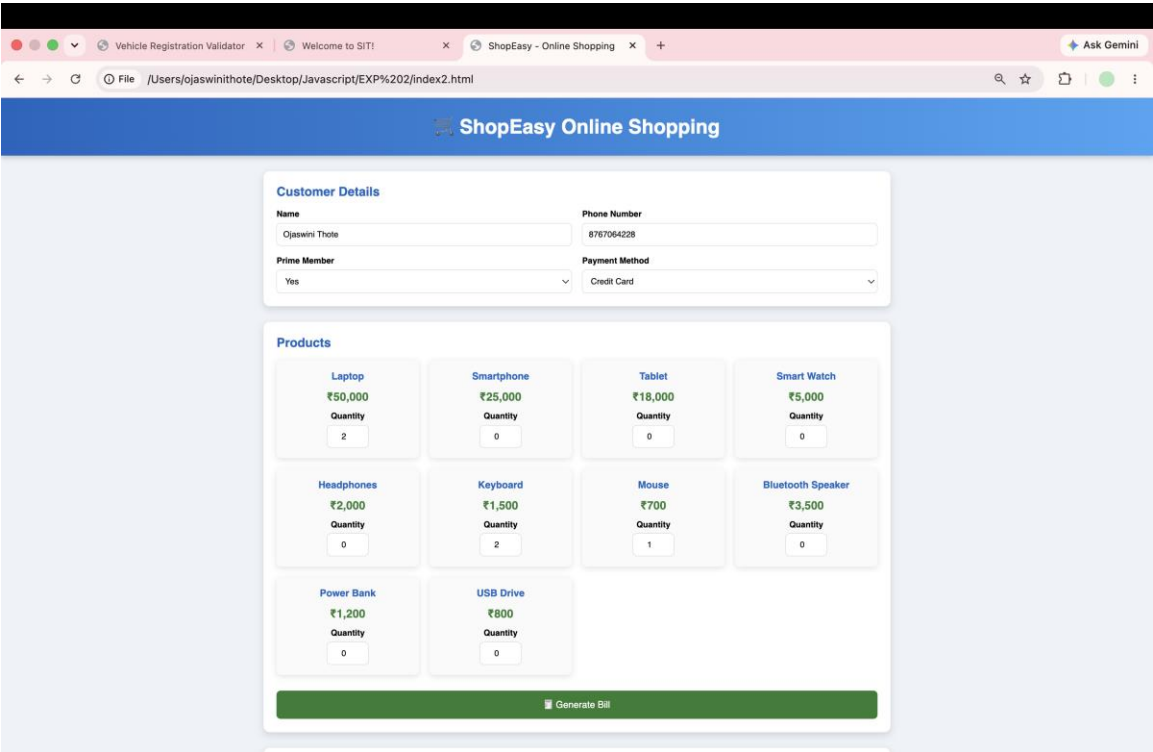
The application displays an online shopping page containing multiple electronic products.

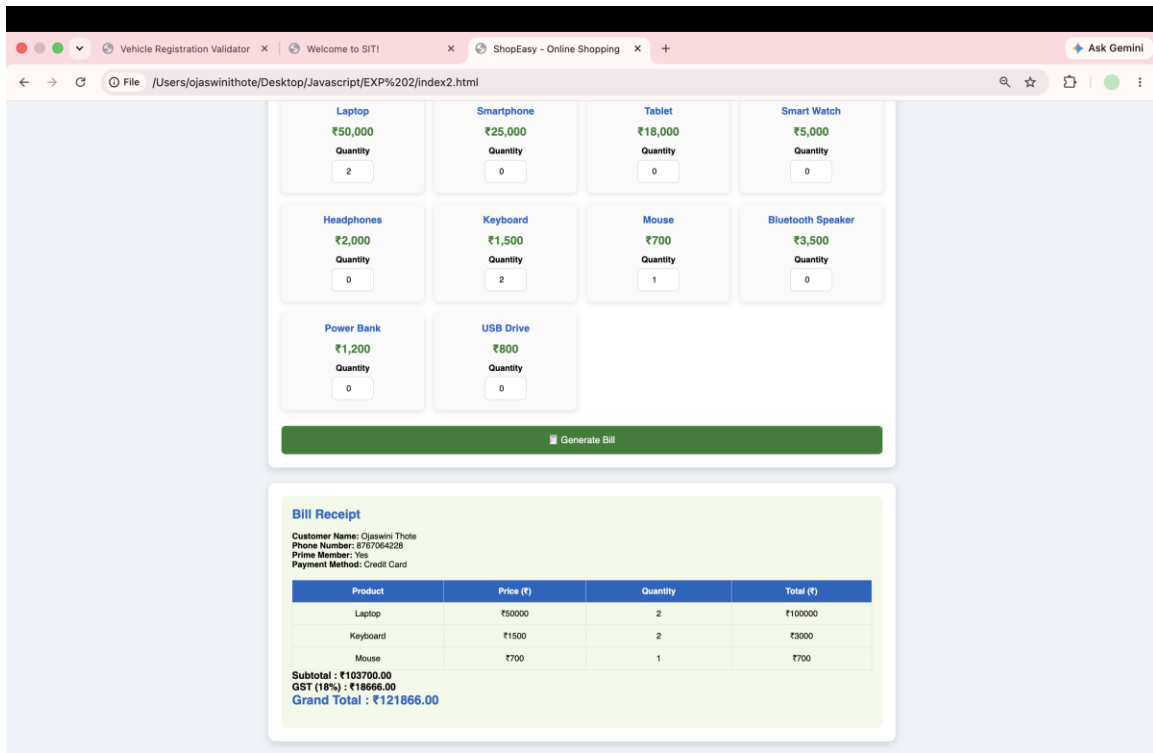
After entering customer details and selecting product quantities, clicking **Generate Bill** displays:

- Customer Information
- Selected Products
- Quantity Purchased
- Product-wise Total
- Subtotal
- GST (18%)
- Grand Total

The browser console displays billing information for verification.

Screenshot





Case Study Title

Billing Calculator Web Application Using JavaScript, HTML and CSS

Case Study Program Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>ShopEasy - Online Shopping</title>

<style>

*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial, sans-serif;
}

body{
```

```

        background:#f4f4f4;
    }

    header{
        background:#0077cc;
        color:white;
        text-align:center;
        padding:20px;
    }

    .container{
        width:90%;
        max-width:1100px;
        margin:30px auto;
    }

    .section{
        background:white;
        padding:20px;
        margin-bottom:20px;
        border-radius:10px;
        box-shadow:0 2px 10px rgba(0,0,0,.2);
    }

    .section h2{
        color:#0077cc;
        margin-bottom:15px;
    }

    label{
        display:block;
        margin-top:12px;
        font-weight:bold;
    }

    input,select{
        width:100%;
        padding:10px;
        margin-top:5px;
        border:1px solid #ccc;
        border-radius:5px;
    }

    table{
        width:100%;
        border-collapse:collapse;
        margin-top:15px;
    }

    th{
        background:#0077cc;
        color:white;
    }

    th,td{
        border:1px solid #ddd;
        padding:10px;
        text-align:center;
    }

    .qty{
        width:70px;
        text-align:center;
    }

```

```
button{
  width:100%;
  margin-top:20px;
  padding:15px;
  background:#28a745;
  color:white;
  border:none;
  border-radius:5px;
  font-size:18px;
  cursor:pointer;
}

button:hover{
  background:#1e7e34;
}

#bill{
  display:none;
  background:#eef8ee;
  padding:20px;
  border-radius:10px;
}

#bill table{
  margin-top:15px;
}

footer{
  background:#222;
  color:white;
  text-align:center;
  padding:15px;
  margin-top:40px;
}

</style>

</head>

<body>

<header>
<h1>📦 Online Shopping</h1>

</header>

<div class="container">

<div class="section">

<h2>Customer Details</h2>

<label>Name</label>
<input type="text" id="name" placeholder="Enter Customer Name">

<label>Phone Number</label>
<input type="text" id="phone" placeholder="Enter Phone Number">

<label>Prime Member</label>

<select id="member">
<option value="true">Yes</option>
<option value="false">No</option>
```

```
</select>

</div>

<div class="section">

<h2>Products</h2>

<table>

<tr>
<th>Product</th>
<th>Price (₹)</th>
<th>Quantity</th>
</tr>

<tr>
<td>Laptop</td>
<td>50000</td>
<td><input type="number" class="qty" data-product="Laptop" data-price="50000" value="0" min="0"></td>
</tr>

<tr>
<td>Smartphone</td>
<td>25000</td>
<td><input type="number" class="qty" data-product="Smartphone" data-price="25000" value="0" min="0"></td>
</tr>

<tr>
<td>Tablet</td>
<td>18000</td>
<td><input type="number" class="qty" data-product="Tablet" data-price="18000" value="0" min="0"></td>
</tr>

<tr>
<td>Smart Watch</td>
<td>5000</td>
<td><input type="number" class="qty" data-product="Smart Watch" data-price="5000" value="0" min="0"></td>
</tr>

<tr>
<td>Headphones</td>
<td>2000</td>
<td><input type="number" class="qty" data-product="Headphones" data-price="2000" value="0" min="0"></td>
</tr>

<tr>
<td>Bluetooth Speaker</td>
<td>3500</td>
<td><input type="number" class="qty" data-product="Bluetooth Speaker" data-price="3500" value="0" min="0"></td>
</tr>

<tr>
<td>Keyboard</td>
<td>1500</td>
<td><input type="number" class="qty" data-product="Keyboard" data-price="1500" value="0" min="0"></td>
</tr>

<tr>
```



```

<td>Mouse</td>
<td>700</td>
<td><input type="number" class="qty" data-product="Mouse" data-price="700" value="0" min="0"></td>
</tr>

<tr>
<td>USB Drive</td>
<td>800</td>
<td><input type="number" class="qty" data-product="USB Drive" data-price="800" value="0"
min="0"></td>
</tr>

<tr>
<td>Power Bank</td>
<td>1200</td>
<td><input type="number" class="qty" data-product="Power Bank" data-price="1200" value="0"
min="0"></td>
</tr>

</table>

<button onclick="generateBill()">
  ☐ Generate Bill
</button>

</div>

<div class="section">

<div id="bill">

<h2>Bill Receipt</h2>

<p><strong>Customer:</strong> <span id="customer"></span></p>
<p><strong>Phone:</strong> <span id="mobile"></span></p>
<p><strong>Prime Member:</strong> <span id="prime"></span></p>

<table>

<thead>

<tr>
<th>Product</th>
<th>Price</th>
<th>Qty</th>
<th>Total</th>
</tr>

</thead>

<tbody id="billItems">

</tbody>

</table>

<h3 id="subtotal"></h3>
<h3 id="gst"></h3>
<h2 id="grandTotal"></h2>

</div>

</div>

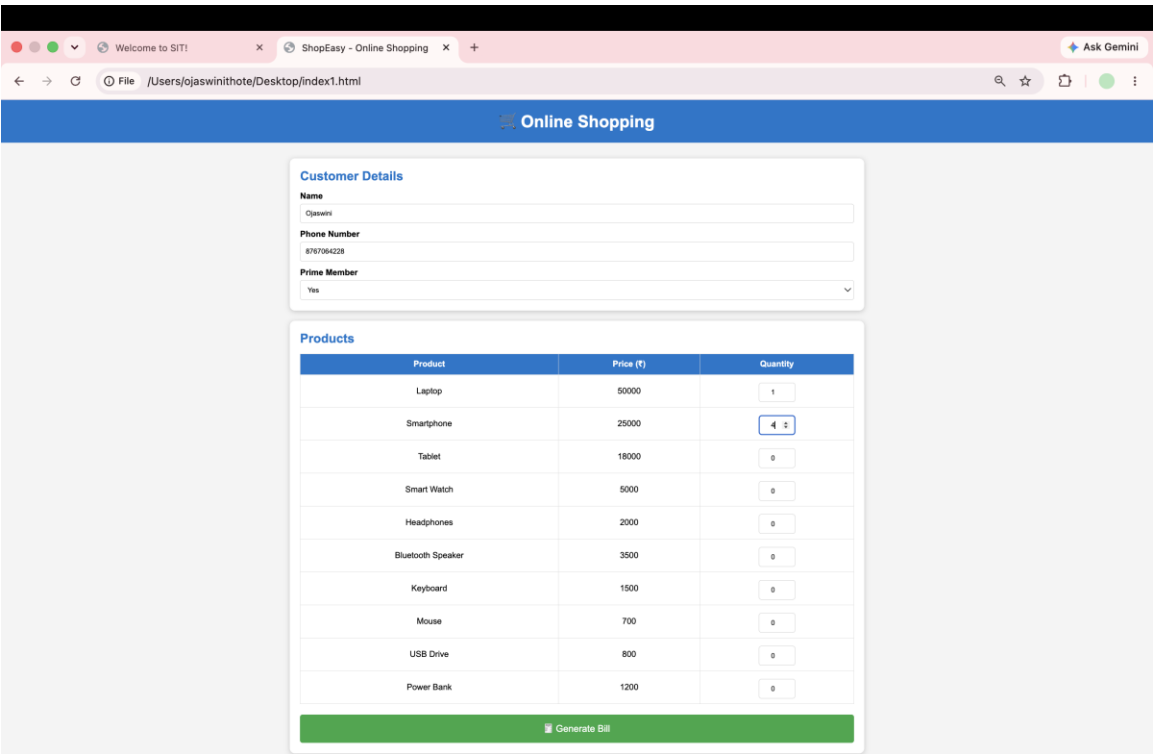
```

</div>

<script src="script2.js"></script>

</body>

</html>



The billing application demonstrates the implementation of JavaScript ES6 concepts.

Concept	Usage
<code>**var**</code>	Used for cart data, loop variables and subtotal calculations
<code>**let**</code>	Used for block-scoped variables during calculations
<code>**const**</code>	Used for GST amount, Grand Total and constant values
<code>**Template Literals**</code>	Used to dynamically generate the invoice using backticks and <code>`\${}`</code> expressions
<code>**Destructuring**</code>	Used to extract product information while preparing the bill

Output Summary

- Customer details are accepted successfully.
- Multiple products can be selected with different quantities.

- GST is calculated automatically.
- Grand Total is displayed in a formatted invoice.
- Billing information is logged in the browser console.

Result / Conclusion

The experiment was performed successfully. The concepts of **var**, **let**, **const**, **template literals**, and **destructuring** were implemented effectively using JavaScript. An online shopping billing calculator was developed that accepts customer details, calculates product totals, applies GST, generates a professional invoice, and dynamically displays the final bill. This experiment demonstrates the practical application of modern JavaScript (ES6) features in developing an interactive web application.